TranStyler: Diffusion-Free 3D Texture Stylization via Disentangled Style and UV Supervision

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Abstract

Text-driven stylization lets an existing 3D asset adopt a new artistic appearance without changing its geometry, but applying image-space objectives directly to a UV map often corrupts seams and UV-island boundaries. The difficulty arises because UV maps combine a distribution shift from natural images with discontinuous charts and non-uniform texel sampling, while paired stylized UV maps are unavailable for supervised training. We present TranStyler, a diffusion-free and paired-data-free framework that disentangles prompt-specific style learning from reusable UV supervision. A Style Optimizer learns the requested appearance online with vision-language guidance, without diffusion denoising. A Geometric Coherence Adapter and a Domain Continuity Adapter, containing 615.27K trainable parameters in total, are trained sequentially offline to preserve boundary structure and UV-domain continuity, then frozen as UV priors. This separation allows the style component to adapt without repeatedly relearning UV-specific constraints. Under a unified evaluation on 11 meshes with 24 rendered views per object, TranStyler outperforms six baselines on all six automatic metrics. It achieves an LPIPS of 0.0952, cross-view CLIP consistency of 0.9259, and an Aesthetic score of 4.832. A study with 61 participants also gives TranStyler the highest ratings, with 4.14 for visual quality and 4.05 for text fidelity. These results show that disentangling style and UV supervision provides an effective diffusion-free route to expressive stylization with structural fidelity.

Figure 1: Examples of text-driven 3D asset re-stylization with TranStyler. Across different objects and prompts, the method changes surface appearance while retaining the input geometry and UV layout.

Introduction

Texturing is a necessary step between 3D asset generation and production use. Mesh-based assets typically store appearance in a UV texture map, a two-dimensional parameterization of the surface that artists can inspect and edit directly. Although recent advances have improved 3D generation and rendering through neural fields, diffusion distillation, and Gaussian splatting (Mildenhall et al. 2021; Poole et al. 2022; Chen et al. 2023b; Wang et al. 2023; Yi et al. 2024; Tang et al. 2023; Kerbl et al. 2023), production pipelines also need to *re-style* an existing asset while retaining its geometry and editable UV layout. This setting supports practical tasks such as creating multiple skins for a game asset, exploring product-design variants, and maintaining a shared mesh across shots. Existing text-driven systems primarily generate or inpaint textures with diffusion priors (Richardson et al. 2023; Chen et al. 2023a; Zeng et al. 2024; Liu et al. 2024; Zhang et al. 2024). They produce strong results for texture synthesis, but their generative machinery is not necessary for every re-stylization task. An alternative is to learn the requested appearance directly while explicitly preserving the input asset's UV structure.

UV maps, however, are not ordinary images. Unwrapping a surface creates seam discontinuities, disconnected island charts, background padding, and non-uniform texel density. Image-space objectives do not model these properties: two adjacent pixels can belong to unrelated surface regions, whereas two distant pixels can meet at a seam on the mesh. Consequently, optimizing semantic alignment alone can introduce cross-island leakage, irregular boundaries, and discontinuities after rendering. We measure this domain mismatch directly: the FID between natural-image and UV-map crops is 259.09, compared with 2.59 between disjoint natural-image splits. The problem is therefore not simply to obtain a strong style prior, but to reconcile prompt-driven appearance changes with constraints imposed by UV parameterization.

TranStyler addresses this tension with a diffusion-free pipeline that separates supervision according to what must change and what must remain stable. A Style Optimizer, abbreviated as SO, adapts online to each text prompt in natural-

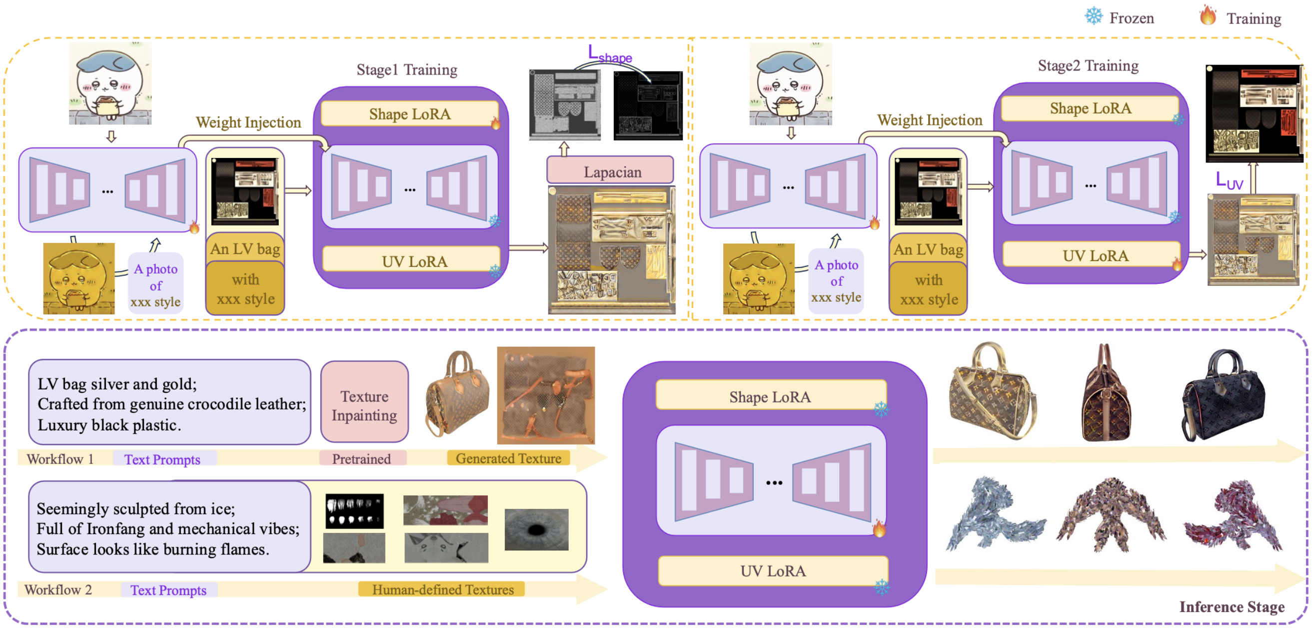
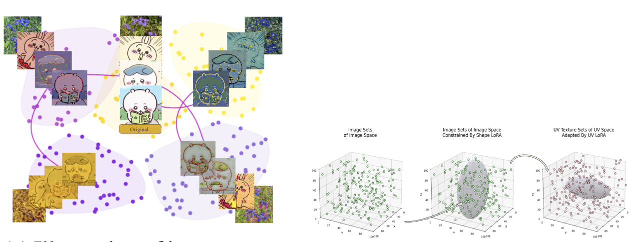


Figure 2: TranStyler pipeline. In Stage 1, SO first learns a text-conditioned appearance and is then frozen while GCA learns Laplacian boundary supervision. In Stage 2, SO and GCA remain frozen while DCA learns UV-domain regularization. At inference, a new prompt updates only SO. GCA and DCA are reused as fixed UV priors.

image space. Two UV adapters are learned offline and reused at inference. The Geometric Coherence Adapter, or GCA, regularizes UV boundaries through Laplacian supervision. The Domain Continuity Adapter, or DCA, constrains the output toward the UV domain. Sequential training isolates these objectives instead of exposing shared trainable weights to all losses at once. On our 11-mesh evaluation set, this design achieves the best value among six baselines on all six automatic metrics and receives the highest human ratings for visual quality and text fidelity.

Our main contributions are:

- We formulate text-driven 3D asset re-stylization as a balance among semantic alignment, UV-boundary preservation, and domain continuity, and quantify the distribution gap between natural images and UV maps.
- We introduce a diffusion-free, paired-data-free framework with sequential supervision: SO performs semantic style search, GCA preserves boundary structure, and DCA enforces UV-domain continuity.
- We develop a hybrid design that learns reusable UV priors offline while adapting the appearance to each prompt online. The staged schedule avoids simultaneous updates under partially conflicting objectives.
- We curate a dataset of 670 textured meshes with raw, masked, and stylized UV maps and evaluate TranStyler against six baselines using automatic metrics and a 61-participant user study.



(a) Illustration of image space style learning.

(b) Styling simulation.

Figure 3: Conceptual view of TranStyler. (a) SO searches the image-model parameter space for a prompt-aligned appearance. (b) GCA and DCA restrict this search using UV boundary and domain constraints before the result is applied to the UV texture.

Related Work

Image Style Transfer. Neural style transfer began with iterative feature-statistics matching (Gatys, Ecker, and Bethge 2016) and was later accelerated by feed-forward networks (Johnson, Alahi, and Fei-Fei 2016; Huang et al. 2019). Arbitrary-style methods align feature statistics (Li et al. 2017; Huang and Belongie 2017; Li et al. 2018) or use attention to establish more flexible correspondences between content and style (Park and Lee 2019; Yao et al. 2019; Xu et al. 2021). Language-guided methods instead use vision-language models to steer a generator or optimize an output directly, as in StyleCLIP, StyleGAN-NADA, and CLIPstyler (Patashnik et al. 2021; Gal et al. 2022; Kwon and Ye 2022). Recent approaches improve spatial control or output quality through region-aware editing, perceptually aligned CLIP gradients, and diffusion priors (Abdal et al. 2022; Ganz and Elad 2024;

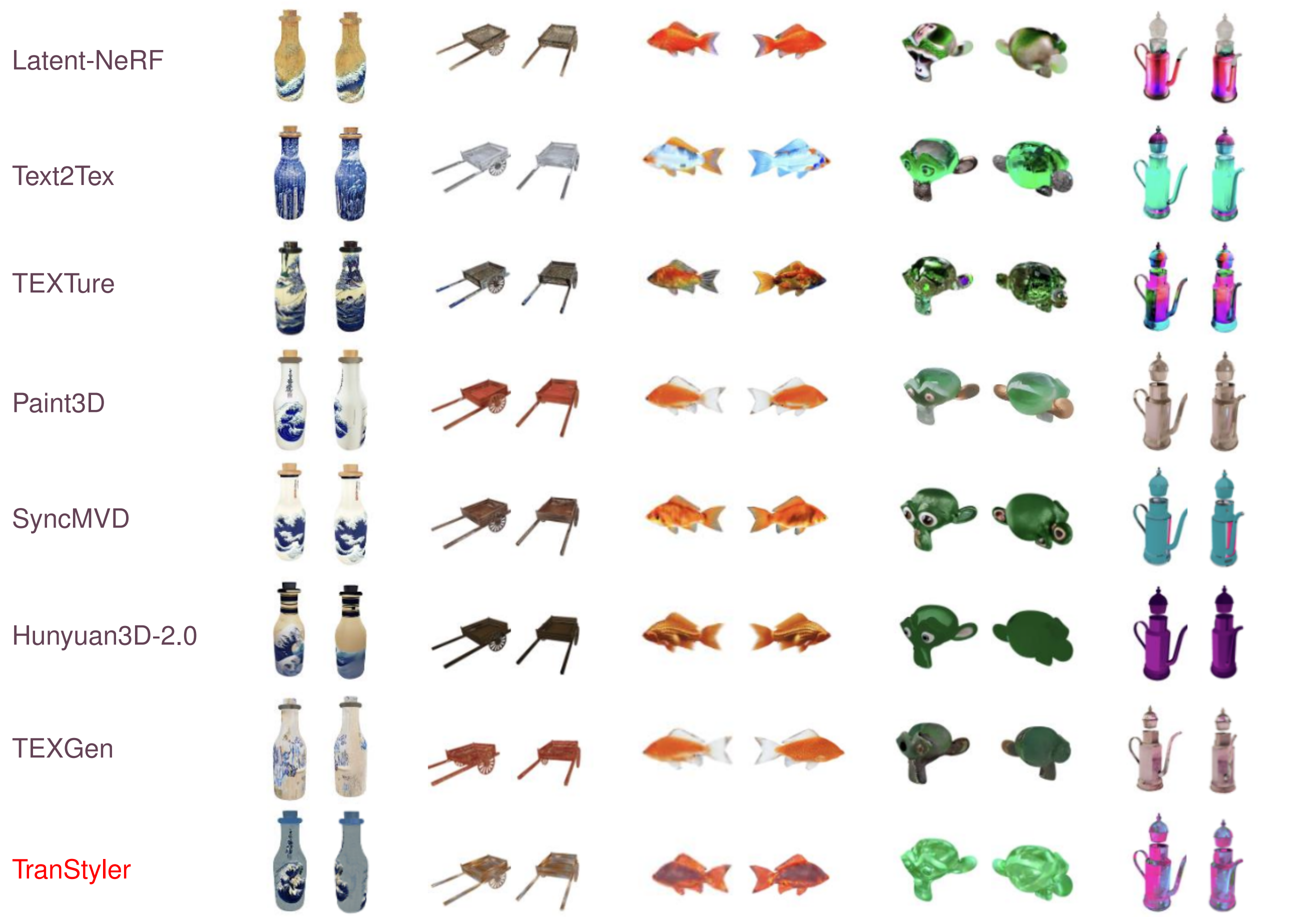


Figure 4: Qualitative comparison on text-driven 3D asset re-stylization. Rows show Latent-NeRF, Text2Tex, TEXTure, Paint3D, SyncMVD, Hunyuan3D-2.0, TEXGen, and **TranStyler**. Each object is rendered from two viewpoints. From left to right, the columns show a Great Wave bottle, an engraved-metal cart, a fire-skin goldfish, a crystal turtle, and a neon kettle. TEXGen is included in this qualitative panel; Table 1 reports the six baselines evaluated under the complete automatic-metric and user-study protocol.

Chung, Hyun, and Heo 2024; Huang et al. 2024; Wang et al. 2024; Zhang et al. 2025). These methods establish strong image-space stylization priors, but their objectives do not account for disconnected UV islands, mesh seams, or non-uniform surface parameterization. TranStyler complements this line of work by retaining text-guided image-space optimization while adding UV-specific structural and domain constraints.

3D Texture Synthesis. Text-guided 3D appearance synthesis has progressed from CLIP-supervised mesh optimization to diffusion-based texture generation. Text2Mesh and CLIP-Mesh optimize appearance, geometry, or normals through differentiable rendering (Michel et al. 2022; Mohammad Khalid et al. 2022), while StyleAvatar specializes text-guided stylization for human avatars (Pérez et al. 2024). Diffusion-based systems such as TEXTure, TexFusion, and Text2Tex iteratively paint a mesh from rendered views (Richardson et al. 2023; Cao et al. 2023; Chen et al. 2023a). Other pipelines synthesize shape and appearance jointly (Siddiqui et al. 2022; Bokhovkin, Tulsiani, and Dai 2023; Poole et al. 2022; Metzer et al. 2023). Paint3D reduces lighting artifacts through UV inpainting and de-lighting (Zeng et al. 2024), whereas SyncMVD improves multi-view consistency by synchronizing the denoising process (Liu et al. 2024). Feed-forward and large-scale diffusion systems, including TEXGen, GenesisTex, and Hunyuan3D-2.0, further improve coverage and fidelity (Yu et al. 2024; Gao et al. 2024; Zhao et al. 2025). These methods primarily address texture creation from text or rendered views. Our setting instead starts from an existing UV-textured asset and focuses on changing its artistic appearance while preserving the editable UV organization.

Test-Time Optimization and Efficient UV Priors. Test-time optimization adapts a model to a particular input or distribution during inference (Sun et al. 2020; Wang et al. 2021). This flexibility is useful in style transfer because each prompt specifies a different appearance. Parameter-efficient mechanisms such as low-rank adaptation can store reusable changes with a small parameter budget (Hu et al. 2022). TranStyler uses this mechanism to realize its UV adapters, but the method's central distinction is supervisory: online optimization captures prompt-specific style, while sequential offline stages encode UV-boundary and domain regularization. This division retains the flexibility of test-time optimization without requiring UV-specific constraints to be relearned for every prompt.

Methodology

TranStyler decomposes re-stylization into three diffusion-free stages, illustrated in Figure 2. The Style Optimizer learns prompt-specific appearance, the Geometric Coherence Adapter preserves UV boundary structure, and the Domain Continuity Adapter regularizes the output toward the UV-texture domain. The two adapters are trained sequentially offline and frozen at inference. Only the Style Optimizer is updated for a new prompt.

Problem Formulation

Let $R \in \mathbb{R}^{H\times W\times 3}$ denote an input UV texture, let T denote a text prompt, and let $f_{\boldsymbol{\theta}}$ denote the stylization network with parameters $\boldsymbol{\theta}$. The output $R_{\text{sty}} = f_{\boldsymbol{\theta}}(R, T)$ should reflect the semantics of T while preserving UV boundaries and remaining coherent in the UV domain. These requirements can be summarized by

$$\begin{aligned}\min_{\boldsymbol{\theta}} \quad & \mathcal{L}_{\text{sem}}(R_{\text{sty}}, T) \\ & + \lambda_{\text{geo}}\mathcal{L}_{\text{geo}}(R_{\text{sty}}, R) \\ & + \lambda_{\text{dom}}\mathcal{L}_{\text{dom}}(R_{\text{sty}}, R),\end{aligned}$$

where $\mathcal{L}_{\text{sem}}$ measures prompt alignment, $\mathcal{L}_{\text{geo}}$ penalizes changes to Laplacian edge structure, and $\mathcal{L}_{\text{dom}}$ penalizes deviation from the input UV distribution. This expression describes the desired solution. TranStyler does not optimize all three terms over the same parameters at once.

Motivation for staged optimization. Semantic alignment encourages visible appearance changes, whereas the geometric and domain terms constrain those changes. Their updates can therefore interfere when they act on shared trainable weights (Yu et al. 2020). On 20 held-out pairs of assets and prompts, the cosine similarity between semantic and geometric gradients is 0.07 ± 0.28, with 30% of measurements below zero. The joint baseline also yields worse FID and Aesthetic scores than sequential training, although its LPIPS is slightly lower, as shown in Table 2. These observations motivate isolating the three objectives rather than assuming that one weighted sum is uniformly optimal.

Modular decomposition. We assign each objective to a dedicated component and freeze earlier components before training the next:

$$\begin{aligned}\text{SO:} \quad & \text{online prompt-specific optimization under } \mathcal{L}_{\text{sem}}, \\ \text{GCA:} \quad & \text{offline geometric regularization under } \mathcal{L}_{\text{geo}}, \\ \text{DCA:} \quad & \text{offline UV-domain regularization under } \mathcal{L}_{\text{dom}}.\end{aligned}$$

At inference, GCA and DCA remain frozen while SO is optimized for the requested prompt. The adapters can therefore be reused when the target style changes.

Style Optimizer: Per-Prompt Online Learning

SO learns the prompt-specific appearance in natural-image space before the UV-specific adapters are applied. Given a natural image Y and a styling prompt T, the CNN encoder-decoder produces $Y_{\text{sty}} = f_{\boldsymbol{\theta}_{\text{SO}}}(Y, T)$. Optimizing this compact network with vision-language guidance requires neither a diffusion model nor paired stylized UV textures and keeps the style component independent of a fixed set of training prompts:

$$\boldsymbol{\theta}^*_{\text{SO}} = \arg\min_{\boldsymbol{\theta}_{\text{SO}}} \mathcal{L}_{\text{SO}}(f_{\boldsymbol{\theta}_{\text{SO}}}(Y, T), Y, T)\,.$$

The SO objective combines five terms. Directional CLIP loss $\mathcal{L}_{\text{dir}}$ aligns the global embedding change with the prompt, and patch loss $\mathcal{L}_{\text{patch}}$ encourages local semantic alignment. Content loss $\mathcal{L}_c$ retains deep features of the input, while total variation $\mathcal{L}_{\text{tv}}$ suppresses high-frequency noise. Finally, an aesthetic term $\mathcal{L}_{\text{aes}}$, implemented with a LAION-trained predictor (Schuhmann et al. 2022), scores differentiably rendered views of the current texture. The complete objective is

$$\begin{aligned}\mathcal{L}_{\text{SO}} = {} & \lambda_{\text{dir}}\mathcal{L}_{\text{dir}} + \lambda_{\text{patch}}\mathcal{L}_{\text{patch}} + \lambda_c\mathcal{L}_c \\ & + \lambda_{\text{tv}}\mathcal{L}_{\text{tv}} + \lambda_{\text{aes}}\mathcal{L}_{\text{aes}},\end{aligned}$$

where each λ weights its corresponding loss.

Disentangled UV Supervision: GCA and DCA

Image-space optimization alone does not encode seam placement, UV-island boundaries, or non-uniform texel density. We therefore treat natural-image appearance as the source domain and UV textures as the target domain. GCA and DCA form a learned bridge between them: GCA constrains local boundary structure, and DCA constrains the resulting texture toward the target UV distribution.

We implement each UV adapter as a low-rank weight increment applied to selected matrices in the SO backbone. This parameterization makes the reusable priors compact; the contribution lies in the separation and scheduling of style and UV supervision rather than in low-rank adaptation itself. Let $\mathbf{W}_{\text{SO}}$ denote an adapted SO weight matrix and let $\Delta\mathbf{W}_{\text{GCA}}$ and $\Delta\mathbf{W}_{\text{DCA}}$ denote the corresponding adapter increments. The effective weight is

$$\mathbf{W}_{\text{full}} = \mathbf{W}_{\text{SO}} + \Delta\mathbf{W}_{\text{GCA}} + \Delta\mathbf{W}_{\text{DCA}}.$$

Across all adapted layers, GCA and DCA contain 615.27K trainable parameters in total.

Stage 1: Offline Geometric Coherence Adapter

Stage 1 preserves editable UV-island boundaries without forcing pixel-identical reconstruction. Let M_{sty} and M_{in} be edge maps obtained by applying a Laplacian kernel to the stylized and input UV textures, respectively. With SO frozen, we train GCA using

$$\mathcal{L}_{\text{GCA}} = \mathcal{L}_{\text{SO}} + \sum_{i,j} |M_{\text{sty}}(i,j) - M_{\text{in}}(i,j)|\,.$$

The Laplacian term penalizes displaced or blurred boundaries while $\mathcal{L}_{\text{SO}}$ retains the learned style signal.

Table 1: Quantitative comparison on 11 meshes with 24 views per object. The user-study columns report mean ratings on a scale from 1 to 5 from 61 participants. FID_{CLIP} is computed against original renders. Best values are in **bold**.

	Automatic Metrics						User Study	
Method	$\text{CLIP}_c\uparrow$	Aesthetic↑	PSNR↑	SSIM↑	LPIPS↓	$\text{FID}_{\text{CLIP}}\downarrow$	Visual↑	Text Fid.↑
Latent-NeRF (Metzer et al. 2023)	0.9217	4.456	18.92	0.8814	0.1091	0.1956	3.36	3.66
Text2Tex (Chen et al. 2023a)	0.9051	4.658	18.38	0.8698	0.1108	0.1835	3.45	3.79
TEXTure (Richardson et al. 2023)	0.9127	4.652	18.63	0.8769	0.1091	0.1956	3.72	4.02
Paint3D (Zeng et al. 2024)	0.9024	4.714	20.64	0.8942	0.0986	0.1975	3.58	3.77
SyncMVD (Liu et al. 2024)	0.9131	4.630	18.92	0.8830	0.1082	0.2281	4.00	4.02
Hunyuan3D-2.0 (Zhao et al. 2025)	0.9089	4.229	16.65	0.8727	0.1166	0.2107	3.89	3.82
TranStyler	**0.9259**	**4.832**	**21.43**	**0.8944**	**0.0952**	**0.1786**	**4.14**	**4.05**

Stage 2: Offline Domain Continuity Adapter

Stage 2 regularizes the output toward the UV-texture domain after boundary structure has been learned. We freeze $\boldsymbol{\theta}_{\text{SO}}$ and GCA, then optimize DCA with an image-space reconstruction term:

$$\mathcal{L}_{\text{DCA}} = \mathcal{L}_{\text{SO}} + \sum_{i,j} |R_{\text{sty}}(i,j) - R(i,j)| \, .$$

This term discourages the style update from erasing the spatial organization of the input UV map. At inference, only $\boldsymbol{\theta}_{\text{SO}}$ changes with the prompt. Both adapter increments remain fixed.

Hybrid Inference: Offline Geometric Constraints with Online Style Search

Hybrid inference separates two sources of variation. UV boundary and domain constraints are reusable across prompts, whereas the requested appearance changes with each prompt. We therefore freeze GCA and DCA and optimize only SO at test time. The frozen adapters bias the update toward boundary-preserving, UV-like outputs, while the unfrozen SO retains enough flexibility to search for a new appearance. In our implementation, this per-prompt optimization takes about one minute on a single RTX 4090 GPU.

Relationship to Image-Space Style Transfer

Applying SO directly to a UV map provides the closest image-space baseline for our design. This baseline cannot distinguish UV content from padding or encode which distant texels meet at a mesh seam. GCA and DCA add these missing constraints without replacing the image-space style prior. The ablations in Section compare SO-only output with each stage and show how the adapters affect boundaries and local continuity.

Experiments

Implementation Details

Data. We construct the TranStyler dataset from textured meshes in Objaverse (Deitke et al. 2023), CAD models in ModelNet40 (Wu et al. 2015), and synthetic textures. After preprocessing, the dataset contains 410 Objaverse textures, 120 ModelNet40 textures, and 140 synthetic textures, for 670 examples in total. For each texture, we derive masks that expose UV boundaries and local details and generate the styled images used for adapter training.

Training and inference. GCA and DCA together contain 615.27K parameters. Stage 1 is trained for 200K steps in approximately 10 hours, and Stage 2 is trained for 100K steps in approximately 5 hours, using 16 RTX 4090 GPUs. At inference, prompt-specific SO optimization takes about one minute on a single RTX 4090 GPU.

Task and protocol. We evaluate text-driven re-stylization on 11 meshes, rendering 24 views per object. Every method receives the same mesh and prompt. The compared systems are designed primarily for texture generation and do not all consume a reference UV texture, whereas TranStyler starts from the asset's UV map. The comparison should therefore be interpreted as output quality under a common asset-and-prompt protocol rather than identical optimization inputs.

Metrics. Cross-view CLIP consistency, denoted by CLIP_c, measures semantic agreement among rendered views (Hessel et al. 2021). PSNR, SSIM, and LPIPS (Zhang et al. 2018) measure similarity to the original renders, while the Aesthetic score measures predicted visual quality. We additionally compute FID_{CLIP} between stylized and original renders (Heusel et al. 2017). This metric is the Fréchet distance in the feature space of CLIP ViT-B/32. Together, the metrics capture view consistency, source preservation, and aesthetic quality. None of them directly measures text fidelity. We therefore complement them with a user study in which 61 participants rate visual quality and text fidelity on a scale from 1 to 5.

Main Results

Qualitative comparison. Figure 4 tests five types of appearance change across two views per object. The qualitative panel additionally includes TEXGen, while Table 1 contains the six baselines evaluated with the complete metric protocol. TranStyler preserves recognizable surface regions while reproducing prompt-specific colors and motifs. In particular, it retains the wave pattern on the bottle, the engraved appearance of the cart, and the localized neon details of the kettle across viewpoints.

The baselines reveal complementary failure modes. Latent-NeRF produces noisy or blurred surfaces in several examples. Text2Tex and TEXTure produce smoother textures but lose some fine prompt-specific patterns. Paint3D occasionally introduces face-dependent transitions. SyncMVD and Hunyuan3D-2.0 maintain coherent materials, but their results are often more uniform than the requested artistic appearance. These observations motivate evaluating both structural preservation and style fidelity rather than relying on a single perceptual metric.

Quantitative comparison. Table 1 reports results under the common 11-mesh, 24-view protocol. TranStyler ranks first on all six automatic metrics. It obtains a CLIP$_c$ of 0.9259, an Aesthetic score of 4.832, a PSNR of 21.43 dB, an SSIM of 0.8944, an LPIPS of 0.0952, and an FID$_{\text{CLIP}}$ of 0.1786. The PSNR, SSIM, LPIPS, and FID$_{\text{CLIP}}$ results indicate stronger preservation of the source-render distribution, while CLIP$_c$ and the Aesthetic score indicate coherent and visually appealing outputs. Taken together with the qualitative results, the metrics show that the improved source preservation does not prevent visible artistic changes.

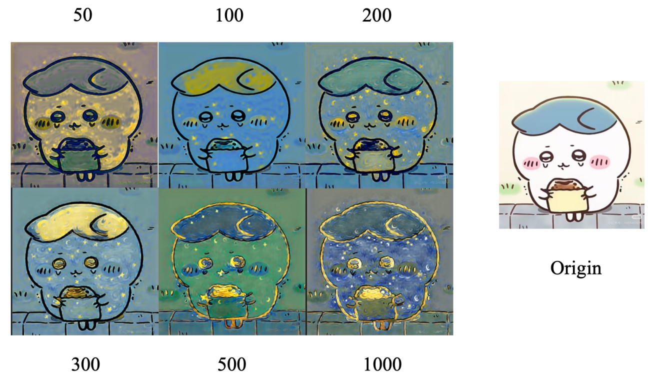
User study. The rightmost columns of Table 1 report mean ratings from 61 participants. TranStyler receives the highest visual-quality rating of 4.14. The next-best result is 4.00. It also obtains the highest text-fidelity rating of 4.05, compared with the next-best value of 4.02. Human ratings therefore support the automatic results while directly evaluating prompt correspondence, which the source-similarity metrics do not capture.

Ablation Studies

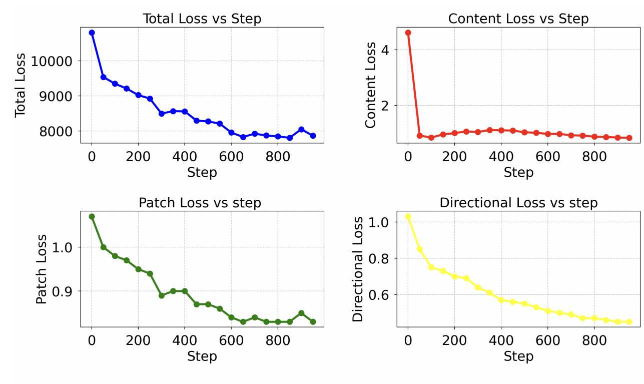
Image Space Online Learning Figure 5 motivates our choice of 200 SO epochs. At epochs 50 and 100, the target style is weak and local details remain blurred. The content loss $\mathcal{L}_c$ stabilizes near epoch 200, when the output exhibits the desired global color and brushwork. Although $\mathcal{L}_{\text{patch}}$ and $\mathcal{L}_{\text{dir}}$ continue to decrease afterward, the global appearance changes little. Instead, later optimization increasingly modifies subject-specific details, such as the eyes, hair, and bread. We therefore use 200 epochs as the operating point that balances prompt alignment with content preservation.

Geometric Coherence Adapter GCA targets the boundary noise produced by SO-only optimization. In Figure 6a, Stage 1 produces sharper UV-island edges in Example 1 and removes local noise in the zoomed regions of Example 2. Figure 6b shows that the corresponding edge loss decreases during training. Together, the visual comparison and optimization curve support the intended role of GCA as a boundary regularizer.

Domain Continuity Adapter DCA targets local discontinuities that remain after image-space optimization. In Example 3 of Figure 7, adding DCA yields smoother transitions than SO alone. The zoom-ins in Example 4 show that it also suppresses isolated spots that do not follow the original UV organization. These examples support the complementary roles of GCA and DCA: GCA regularizes boundaries, whereas DCA improves local UV-domain continuity.



(a) Chiikawa styling results at different epochs for a Van Gogh style prompt.



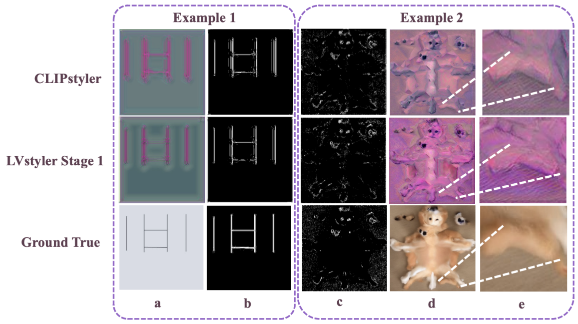
(b) Training losses across SO optimization epochs.

Figure 5: Selecting the number of SO optimization epochs. The top panel presents intermediate outputs for the Chiikawa prompt. The bottom panel reports total, content, patch, and directional losses. Content stabilizes near epoch 200. Later epochs provide limited global change and increasingly alter local subject details.

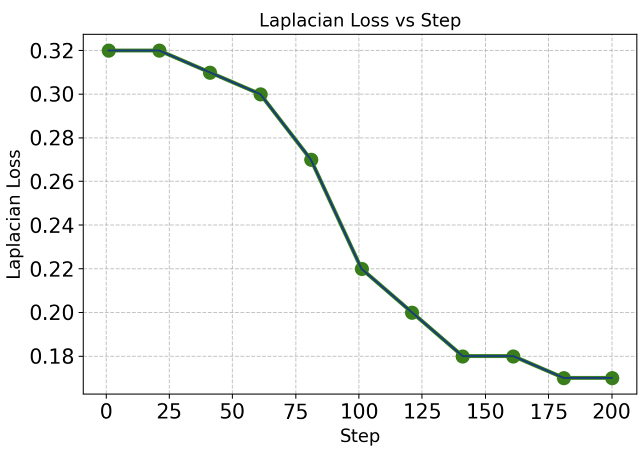
Sequential vs. joint training. Table 2 compares the two schedules on 3D renders. Sequential training improves FID from 114.67 to 111.89 and Aesthetic score from 4.347 to 4.376. Joint training has a slightly lower LPIPS of 0.0822, compared with 0.0833 for sequential training. To examine this trade-off, we measure the cosine similarity between $\nabla\mathcal{L}_{\text{SO}}$ and $\nabla\mathcal{L}_{\text{GCA}}$ on 20 held-out pairs of assets and prompts. The mean is close to zero at 0.07 ± 0.28, but 30% of measurements are negative and the minimum is -0.60. Thus, conflict is episodic rather than universal. Sequential staging prevents these episodes from changing an already trained component. This schedule improves the balance between FID and Aesthetic score without changing the inference architecture.

Adapter rank sensitivity. Table 3 evaluates ranks $\{2, 4, 8, 16\}$. No rank dominates all three metrics, and increasing the rank produces no monotonic improvement. We therefore retain rank 4 as a compact default, giving 615.27K adapter parameters in total.

UV domain shift. We quantify the motivation for DCA



(a) Geometric Coherence Adapter performance.



(b) Laplacian loss during Stage 1 training.

Figure 6: Stage 1 ablation. The top panel shows that GCA produces cleaner UV boundaries and local details than SO alone. The bottom panel shows the decrease in Laplacian edge loss during GCA training.

using 256×256 crops. FID between natural images in the style corpus and UV-map crops from the 670-mesh dataset is 259.09, whereas FID between disjoint natural-image splits is 2.59. The large difference verifies that UV patches and natural images occupy markedly different feature distributions.

Adapter portability. We evaluate the frozen adapters on three ModelNet40-derived objects: Bus, Car, and Suzanne. The adapters are trained predominantly on Objaverse meshes.

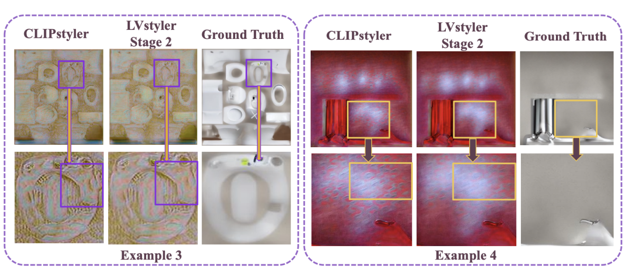


Figure 7: Stage 2 ablation. DCA reduces abrupt transitions and spurious local spots relative to SO-only optimization.

Table 2: Sequential and joint training on 3D renders. Bold values indicate the better result for each metric. Sequential training is our default.

Variant	FID↓	LPIPS↓	Aesthetic↑
Joint training	114.67	**0.0822**	4.347
Sequential	**111.89**	0.0833	**4.376**

Table 3: Adapter rank sensitivity on 3D renders. No rank dominates all metrics. **Bold** denotes the selected default at rank 4.

Variant	FID↓	LPIPS↓	Aesthetic↑
Adapter rank $= 2$	112.94	0.0823	4.384
Adapter rank $= 8$	111.49	0.0834	4.342
Adapter rank $= 16$	114.59	0.0830	4.373
Full, rank $= 4$	**111.89**	**0.0833**	**4.376**

The adapter-augmented model obtains LPIPS 0.1416, compared with 0.1418 for SO only. This small difference shows that adding the adapters does not degrade LPIPS on these cross-category examples, although broader transfer evaluation remains future work.

Metric interpretation. FID_{CLIP} and LPIPS measure distance to the original-render distribution, so they reward source preservation rather than style strength alone. TranStyler leads on both source-similarity metrics and on Aesthetic score, while its text fidelity is supported by the user study. The agreement across these complementary measurements is more informative than any single metric.

Conclusion

TranStyler provides a diffusion-free approach to text-driven stylization of UV-textured 3D assets by disentangling prompt-specific appearance learning from reusable UV supervision. SO adapts online to the requested style, while GCA and DCA are trained offline to regularize UV boundaries and domain continuity. On 11 meshes with 24 views per object, TranStyler ranks first among six baselines on all six automatic metrics. It achieves an LPIPS of 0.0952, a CLIP_c of 0.9259, and an Aesthetic score of 4.832. In the 61-participant study, it also receives the highest ratings, with 4.14 for visual quality and 4.05 for text fidelity. These results support the central design choice: style adaptation and UV preservation are easier to balance when assigned to separately supervised components. The current evaluation is limited to a small test set, and the method still requires per-prompt online optimization. Extending the study to more diverse UV layouts and distilling the online search into a feed-forward model are important directions for future work.

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